

L03 LP Duality and zero-sum games

CS 280 Introduction to Algorithmic Game Theory

Ioannis Panageas

Linear Programming

Problem (Linear Program (Feasibility)). *Suppose we are given a linear program in the standard form*

$$\begin{aligned} Ax &\leq b \\ x &\geq 0. \end{aligned}$$

where A is of size $n \times m$.

Goal: Find a *feasible* solution x^* (if there is one).

Remark: We have n constraints and m variables.

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Lemma (Equivalence). These two problems are polynomial time equivalent.

Linear Programming

Problem (Primal Formulation). *Suppose we are given a linear program in the standard form*

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Remark: We have m constraints and n variables!

Linear Programming

Facts (Four possible cases).

1. Primal and dual both feasible and bounded.
2. Primal feasible and unbounded, dual infeasible.
3. Primal infeasible, dual feasible and unbounded.
4. Primal and dual both infeasible.

Linear Programming

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Let's focus on case 1.

Theorem (**Weak duality**). *Assume that primal and dual are feasible. It holds that*

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Moreover, let $y \in D$. We have that $y^\top Ax \leq y^\top b$.

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Moreover, let $y \in D$. We have that $y^\top Ax \leq y^\top b$.

$$\text{Therefore, } c^\top x \leq y^\top Ax \leq y^\top b.$$

Since x, y were arbitrary it follows $\max_{x \in P} c^\top x \leq \min_{y \in D} b^\top y$.

Linear Programming

Theorem (Strong duality). *Assume that primal is feasible and bounded. It actually holds that*

$$\max_{x \in P} c^\top x = \min_{y \in D} b^\top y$$

Remark: The proof is much harder, it uses Farkas' lemma.

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Example.

Primal

$$\begin{aligned} & \max z \\ \text{s.t. } & 3x_1 - 2x_2 - z \geq 0 \\ & -x_1 + x_2 - z \geq 0 \\ & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

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Example.

Primal

$$\begin{array}{ll} \max & 0 \cdot x_1 + 0 \cdot x_2 + 1 \cdot z \\ \text{s.t.} & \begin{pmatrix} -3 & 2 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} \leq 0 \\ & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{array}$$

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Dual

$$\begin{aligned} & \min 0 \cdot y_1 + 0 \cdot y_2 + 1 \cdot w \\ \text{s.t. } & \begin{pmatrix} -3 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ w \end{pmatrix} \geq 0 \\ & y_1 + y_2 = 1 \\ & y_1, y_2 \geq 0 \end{aligned}$$

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Example. Sol $x_1, x_2 = (\frac{3}{7}, \frac{4}{7}), y_1, y_2 = (\frac{2}{7}, \frac{5}{7}), w = z = \frac{1}{7}$

Primal

$$\begin{aligned} & \max 0 \cdot x_1 + 0 \cdot x_2 + 1 \cdot z \\ \text{s.t. } & \begin{pmatrix} -3 & 2 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} \leq 0 \\ & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

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They match, optimality!!

Linear Programming

Facts (polynomial time).

1. Solving a linear program with rational data is in P .
2. The first polynomial time algorithm was the ellipsoid method (Khachiyan).
3. Interior point methods give polynomial time algorithms and are widely used.
4. Simplex can take exponential time in the worst case, but is often fast in practice.

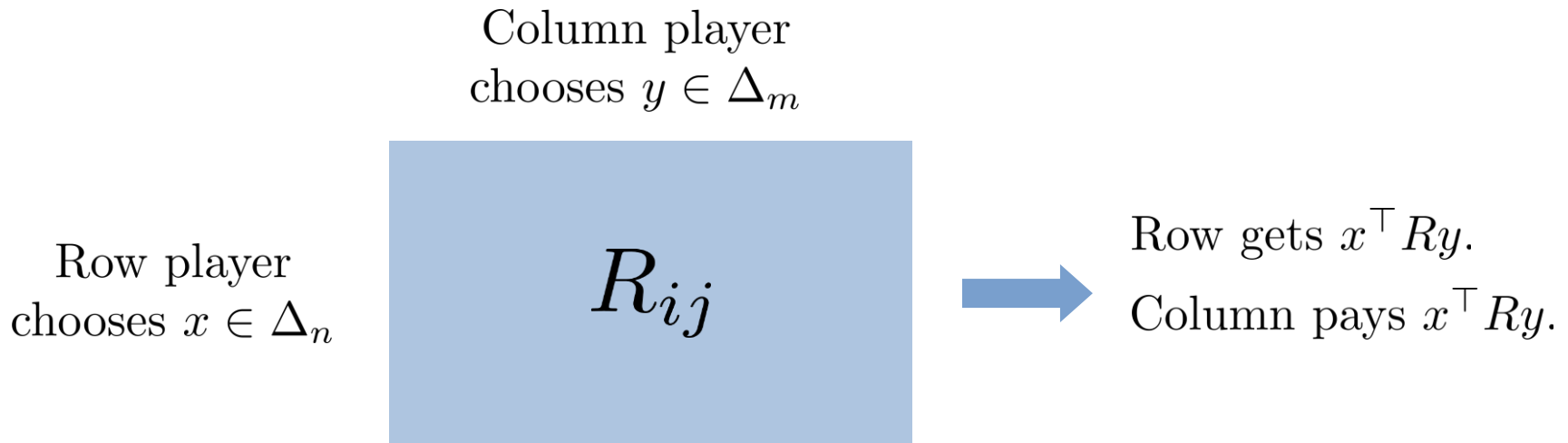
Back to zero-sum Games

Question: Why do we care about LP? Recall the zero-sum game example from the previous lecture!

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Answer: We can formulate the problem of computing Nash in zero-sum using LP!



Zero-sum Games as LPs

Assume player x plays first and wants to get **at least** z . For **all pure strategies** of y , x should get at least z . Formally:

$$x^\top R \geq z \cdot \mathbf{1}^\top$$

Zero-sum Games as LPs

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$$x^\top R \geq z \cdot \mathbf{1}^\top$$

or

$$-x^\top R + z \cdot \mathbf{1}^\top \leq 0$$

Moreover, x should be a randomized strategy. Formally:

$$x^\top \mathbf{1} = 1$$
$$x \geq \mathbf{0}$$

Zero-sum Games as LPs

LP for player x :

$$\begin{aligned} \max \quad & z \\ x^\top R & \geq z \cdot \mathbf{1}^\top \\ x^\top \mathbf{1} & = 1 \\ x & \geq \mathbf{0} \end{aligned}$$

Remark: Notice that the maximum above is the same as

$$\max_{x \in \Delta_n} \min_{y \in \Delta_m} x^\top R y$$

Zero-sum Games as LPs

Consider the dual of the previous LP:

$$\begin{aligned} \min z' \\ -y^\top R^\top + z' \cdot \mathbf{1}^\top &\geq 0 \\ y^\top \mathbf{1} &= 1 \\ y &\geq \mathbf{0} \end{aligned}$$

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Set $z'' = -z'$ the above becomes

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Zero-sum Games as LPs

Consider the dual of the previous LP:

This is the LP as if y player would play first with sign flipped!

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Nash equilibrium and LP

$$\begin{array}{ll} \text{LP1} & \max z \\ & x^\top R \geq z \cdot \mathbf{1}^\top \\ & x^\top \mathbf{1} = 1 \\ & x \geq \mathbf{0} \end{array}$$

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Theorem. Let (x^*, z) be optimal for LP1, and (y^*, z'') be optimal for LP2, then (x^*, y^*) is a Nash equilibrium of the zero sum game with payoff matrix R . The payoff of the row player is z and of the column player is $z'' = -z$.

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$$x^{*\top} R y^* = z!$$

No matter what y does if x chooses x^* , y pays at least z .

No matter what x does if y chooses y^* , x gets at most z .

Thus it is a Nash!

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Proof. Homework!

Corollaries

Theorem (**von Neumann minimax theorem**). *It holds that*

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Theorem (Convexity of Nash Equilibria). *The set of Nash equilibria in a zero-sum game is convex.*